

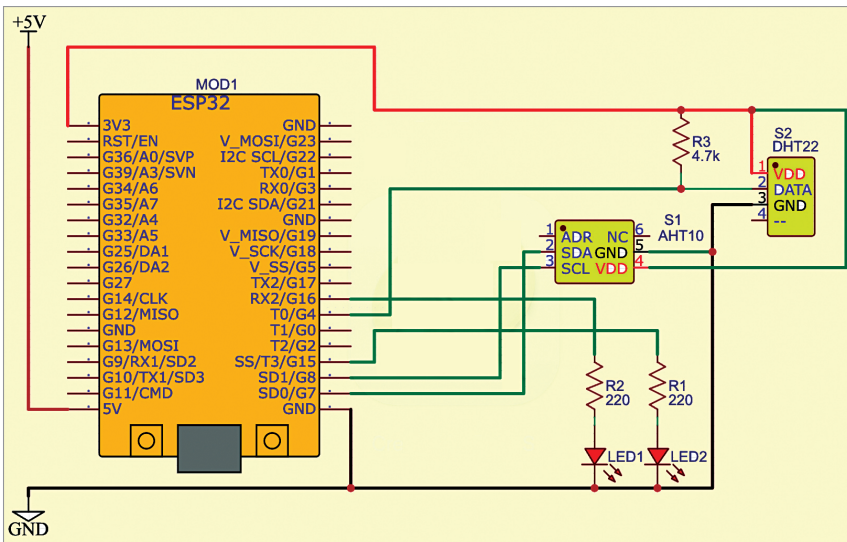


Fig. 3: Circuit diagram

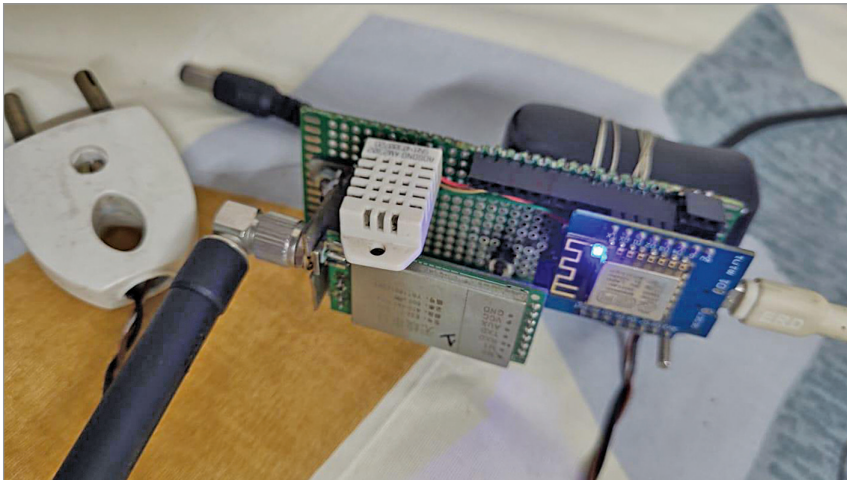


Fig. 4: Prototype with DH22 sensor

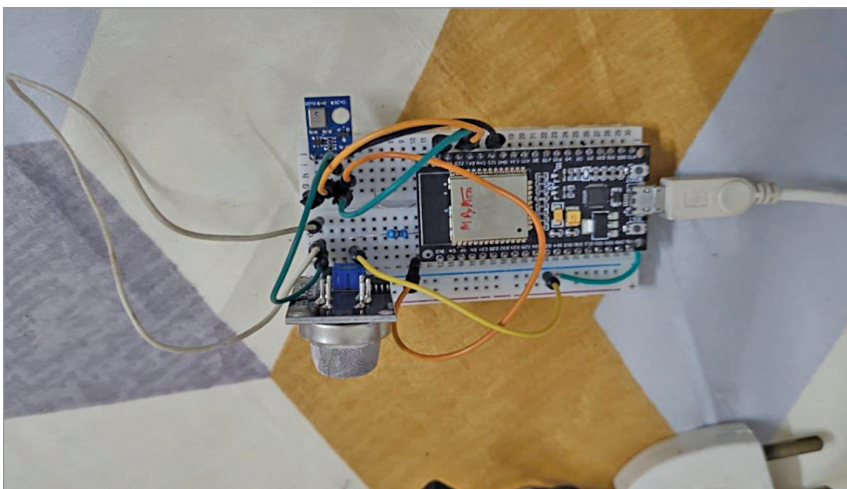


Fig. 5: Prototype with MQ135 sensor

MQ135, and share data globally via MQTT.

This is automation at its core—streamlining control and monitoring for a more intelligent home. The best part is that setting up the ESP32 or ESP8266 is straightforward.

Bluetooth proxy

Bluetooth devices have been part of daily life for years, yet integrating them into home automation has always been challenging. This is one reason Zigbee has remained a preferred choice. However, ESPHome Automation introduces an effective alternative.

Nearby BLE devices automatically connect by setting up an ESP32 as a Bluetooth proxy. The data from these devices becomes visible on the ESPHome dashboard under the 'ESPHome Bluetooth Proxy' device.

It is important to note that the ESPHome Bluetooth Proxy is not designed to connect directly to mobile phones or appear in a phone's Bluetooth settings like a standard Bluetooth device. Instead, it bridges Home Assistant and nearby BLE devices, such as temperature sensors, smart bulbs, fitness bands, or other smart home peripherals.

In Bluetooth proxy mode, the ESP32 does not advertise itself as a connectable device (e.g., a speaker or keyboard). Instead, it listens to nearby BLE devices and silently forwards the data to the Home Assistant. There are no notifications or pairing prompts. You need to check the logs to confirm that it is working—it will notify you when it 'finds a new device.' The simplicity and power of this silent integration make Bluetooth proxy a revolutionary feature in home automation.

The components required for the automation system are listed